



Evaluation of body weight prediction equations and development of morphological models for day-old beef-on-dairy calves

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ABSTRACT

The objectives of this study were to evaluate the accuracy of published equations for estimating BW of beef × dairy (B × D) male calves and to develop, test, evaluate, cross-evaluate, and validate new predictive equations based on morphological measurements of day-old B × D male calves. Three hundred twenty B × D male calves were enrolled upon arrival at the calf-raising facility, and morphological measurements such as BW, heart girth (HG), hip height (HH), and hoof circumference (HC) were recorded from each calf within the first 24 h of life. Forty-six published peer-reviewed equations were evaluated against the development dataset using root mean squared error (RMSE), concordance correlation coefficient (CCC), and mean squared error decomposition (mean bias, slope bias, and residual error) to assess agreement between observed and predicted BW. Moreover, 3 linear equations were developed using a forward selection criterion for HG (Equation 1), HG and HH (Equation 2), and HG, HH, and HC (Equation 3). In addition to linear models, evaluation of quadratic, cubic, and exponential models was assessed to determine the relationship between BW and morphological traits. Among the tested models, the linear equation proved more effective and simpler than the quadratic, cubic, and exponential models and was ultimately selected for validation. Overall, 76% of the published equations demonstrated poor agreement with observed BW, as indicated by an average CCC (0.49) and an average BW RMSE of 107.00 kg. An additional 17% of the published equations achieved moderate agreement, with average CCC values between 0.50 and 0.60 and an RMSE for BW of 6.20 kg; only 7% (the top-performing published equations) had CCC values greater than 0.60 and an RMSE for BW

of 4.50 kg. All 3 developed equations exhibit greater predictive performance than 43 (93%) of the published equations, with an average CCC of 0.75 and an RMSE for BW of 3.50 kg. Cross-evaluation of the 3 developed equations demonstrated moderate to high BW-predicting performance (RMSE = 3.80 kg; CCC = 0.74). The validation assessment using an independent dataset of 176-d-old B × D male calves that only included HG as a morphological measurement indicated that the HG equation (Equation 1) had a moderate to high performance (RMSE = 3.80 kg; CCC = 0.64) in predicting the BW of day-old B × D male calves. Thus, indicating that this developed equation (Equation 1) can be applied as an on-farm tool to predict the BW of day-old B × D male calves.

Key words: calf, modeling, regression, weight

INTRODUCTION

Body weight can be used as a crucial indicator of animal growth performance, health, and management decisions (Putnam et al., 1964; Dikmen et al., 2012; Halachmi et al., 2019). Producers and researchers have developed different methods to measure or predict BW in cattle, with varying accuracy (Dingwell et al., 2006). Traditionally, cattle BW is measured using a digital scale platform that requires the animal to move across it, which can be a time-consuming and labor-intensive process that may negatively affect cattle growth performance (Wigham et al., 2018; Firdaus et al., 2023). Digital scales are not standard on many dairy farms due to their financial cost and potential logistical challenges; furthermore, this infrastructure gap has encouraged previous researchers to develop alternative BW estimation methods (Heinrichs et al., 1992, 2017). Some alternative practices involve predicting BW from morphological traits, such as heart girth (HG), hip height (HH), and hoof circumference (HC). Moreover, previous research by Davis et al. (1961) and Heinrichs et al. (2017) has reported that BW and HG are strongly correlated in dairy calves.

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

Historically, beef semen has been used in dairy farms as a reproductive strategy to reduce replacement heifer inventory and potentially increase farm profitability (Wolfová et al., 2007; Geiger, 2019; Cabrera, 2022). Berry (2021) reported that beef is a byproduct of the dairy industry but can also be one of dairy producers' most significant sources of cash flow. Moreover, calves born from a beef breed sire have been reported to have a greater market value than calves born from dairy breed sires (Dal Zotto et al., 2009; McHugh et al., 2010; Berry et al., 2018). According to the NAAB (2024), domestic beef semen sales have increased by 282.90% (7,186,456 doses) from 2017 to 2024, although domestic dairy semen sales have decreased by 33.27% (7,717,093 doses) during the same period. Although not all of the increase in beef semen sales is going to dairy farms, the similar decrease in dairy semen sales indicates an increase in beef semen adoption by dairy producers, which has been reported by previous surveys conducted in different regions of the country (Pereira et al., 2022; Felix et al., 2023).

Cattle frame size refers to the overall body dimensions of an animal, typically characterized by body length and height (Tatum et al., 1986; Fuerniss et al., 2023). Although purebred dairy cattle generally exhibit greater frame size than age-matched beef calves (Basiel and Felix, 2022), Nolan (2022) and Breneman (2024) reported that one advantage of using beef semen in dairy herds is the production of calves with moderate frame size, which may improve feed efficiency and growth performance. However, early-life morphological differences among dairy, beef, and beef $\times$ dairy (**B $\times$ D**) calves in frame size and body proportions can create structural mismatches with BW prediction equations developed in the purebred dairy population (Heinrichs and Hargrove, 1987; Heinrichs et al., 1992; Wilson et al., 1997). The potential mismatch in BW-predicting equations can be crucial during the first days of life of B $\times$ D calves, as they are often sold and transported, where direct weighing is frequently impractical (Cramer et al., 2024). Early-life BW is also a key predictor of health risks, nutrient requirements, and preweaning growth potential, yet calf managers commonly rely on estimated BW because scales are often unavailable in newborn handling areas (Dingwell et al., 2006; Heinrichs et al., 2017). As the B $\times$ D calf market expands (Berry, 2021; Basiel and Felix, 2022), these previously described biological and logistical considerations underscore the need to develop BW prediction tools calibrated specifically to B $\times$ D calf morphology.

To the author's knowledge, no BW-predicting equations have been developed solely for B $\times$ D calves. Published equations were primarily derived from either purebred dairy calves (Heinrichs et al., 1992, 2017) or beef breed calves (Bahashwan, 2014; Tebug et al., 2016). Therefore,

it is hypothesized that published equations for estimating BW may not be suitable for B $\times$ D male calves. The objectives of this study were to evaluate the accuracy of published equations for estimating BW of B $\times$ D male calves and to develop, test, evaluate, cross-evaluate, and validate new predictive equations based on morphological measurements of day-old B $\times$ D male calves.

MATERIALS AND METHODS

The study was conducted in a commercial calf-raising facility in the San Joaquin Valley, California. Animal handling was approved and conducted in accordance with the Institutional Animal Care and Use Committee (IACUC # 4353). Three hundred twenty B $\times$ D male calves weighing from 31.75 to 61.70 kg (average initial BW of 42.6 ± 5.6 kg), sourced from 14 commercial dairies, were enrolled in the study in their first 24 h of life. The number of calves contributing to the development dataset averages 23 per dairy farm. Once arrived at the calf-raising facility, calves received a 2-mL nasal vaccine for bovine respiratory disease (Inforce 3, Zoetis Animal Health, Kalamazoo, MI) and a 1-mL subcutaneous vaccine for *Salmonella* Dublin (EnterVene-d; Boehringer Ingelheim). Navel from each calf were sprayed with iodine, followed by castration using a rubber ring, and the calves were further allocated to individual calf hutches.

BW and Morphological Measurements

Body weight was recorded on arrival (d 1) using a digital scale (Tru-test digital scale, model S3, Datamars, Mineral Well, TX), followed by morphological measurements taken once per calf after placement in their assigned hutch. Trained personnel ensured calves were standing square on the hutch's base before measurements were collected, once using a measuring tape (Prym 60 inch, Prym Consumer USA Inc.). Heart girth was measured as the chest circumference taken behind the shoulder using a measuring tape. Hip height was calculated as the distance from the top of the hip to the base of the hoof in the hind leg, ensuring the calf was standing completely squared with the hind leg extended to the natural position. Hoof circumference was measured around the coronary band of the front right hoof.

Database, Calculations, and Statistical Analysis

The development database used in the study contained 320 single-time point observations, with one record per male calf for BW, HG, HH, and HC. Therefore, authors recognize that the robustness of the developed models may be limited when applied to populations exhibiting different phenotypic variability.

Table 1. Summary of source studies used in the evaluation of published equations to predict BW

Reference	Method	Morphological measure ¹	n	BW (kg)		Cattle		
				Mean	SD	Type ²	Breed ³	Sex ⁴
Heinrichs et al. (1992)	Tape measure	HG	—	—	—	DA	H	F
Pruitt et al. (1995)	Tape measure	HC	81	40.1	5.7	BE	ASCR	—
Wilson et al. (1997)	Tape measure	HG	789 ⁵	114 ⁵	11.7 ⁵	DA	H	M
Reis et al. (2008)	Tape measure	HG	62	250.8	96.5	DA	MU	M
Ozkaya and Bozkurt (2009)	Tape measure	HG	140	471.7 ⁵	118.6 ⁵	DA	MU	F
Parish et al. (2009)	Spring scale	BW	587	36.1	5.5	BE	MU	B
Yan et al. (2009)	Tape measure	HG	146	574	74.4	DA	H	F
Long et al. (2012)	Tape measure	HC	872	35.6 ⁵	0.2 ⁵	DA	HJ	B
Lesosky et al. (2013)	Tape measure	HG	—	—	—	DA	H	—
Bahashwan (2014)	Tape measure	HG	36	135.6 ⁵	2.6 ⁵	BE	D	B
Heinrichs et al. (2017)	Tape measure	HG	586	275.4 ⁵	—	DA	H	F
Tebug et al. (2016)	Tape measure	HG	140	301.6 ⁵	93.7 ⁵	BE	IZ	F
Cantor et al. (2020)	Partial weight scale	BW	20	35.0	6.0	DA	H	F
Udoh et al. (2021)	Tape measure	HG	86	200.6 ⁵	51.9 ⁵	BO	MU	—
Rocha-Silva et al. (2024)	Tape measure	HG	155	19.9	—	BO	C	—
Silva et al. (2024)	Tape measure	HG and HH	210	55.9 ⁵	15.2 ⁵	DA	H	B

¹HC = hoof circumference; HG = heart girth; HH = hip height.

²DA = dairy animals; BE = beef animals; BO = both dairy and beef animals.

³HJ = Holstein and Jersey; H = Holstein; D = Dhafari; IZ = Indigenous Zebu; C = Curraleiro Pé-Duro; ASCR = Angus, Simmental, and Crossbred; MU = multiple breeds not explicitly defined.

⁴F = female; M = male; B = both female and male.

⁵Values are presented as means calculated from data reported in the published equations. Minor differences from published values reflect averaging across multiple observations to obtain standardized mean estimates.

Descriptive statistics were generated using the *psych* library (Revelle, 2013, 2025), and normality was visually assessed using a histogram (Supplemental Figure S1, see Notes) produced with the *summarytools* package (Comtois and Comtois, 2025). The independence of residuals was assessed using the Durbin–Watson test, and homoscedasticity was evaluated using the Breusch–Pagan test from the *lmtest* package (Zeileis and Hothorn et al., 2002). All statistical procedures were conducted in R (R Core Team, 2024) using RStudio (v.2024.12.1.563). Statistical significance was declared at $P \leq 0.05$, and tendencies were declared at $0.05 < P < 0.10$. Model performance was assessed according to Tedeschi (2006) by using the concordance correlation coefficient (CCC), root mean square error (RMSE) and mean square error (MSE). Details regarding model development, test, evaluation, cross-evaluation, and external validation are presented in the following sections.

Peer-Reviewed Published Equations

A structured search retrieved published beef and dairy equations predicting BW, resulting in 46 equations that were compiled into a database. Equations were screened using predefined selection criteria to ensure alignment with the study objectives. The selection criteria required that (1) equations included key variables of interest (i.e., BW, HG, HH, HC), and (2) were developed for calves and heifers. The criteria ensured compatibility between pub-

lished equation inputs and the measurements available in the development dataset. All evaluated studies and their summary statistics are presented in Table 1. Equation description and applied criteria are provided in Table 2. The predictive ability of the published equations for the BW developed by Heinrichs et al. (1992), Pruitt et al. (1995), Wilson et al. (1997), Reis et al. (2008), Ozkaya and Bozkurt (2009), Parish et al. (2009), Lesosky et al. (2013), Long et al. (2012), Yan et al. (2009), Bahashwan (2014), Tebug et al. (2016), Heinrichs et al. (2017), Cantor et al. (2020), Udoh et al. (2021), Rocha-Silva et al. (2024), and Silva et al. (2024) were evaluated using d1 observations from the development database (Table 3).

Model evaluation was performed using the *lme4* and *epiR* packages (Bates et al., 2015; Stevenson and Sergeant, 2025). Assessment variables included BW observed (kg) and predicted mean (kg), adjusted R^2 , Akaike information criterion (AIC), log-likelihood, RSME, MSE bias (% of MSE and kg), slope bias (% of MSE and kg), residual error (% MSE), and CCC (Tedeschi, 2006). Each model was evaluated using a structured approach: assessing the overall magnitude of prediction error (RMSE), followed by goodness-of-fit and agreement metrics (CCC, adjusted R^2), and MSE decomposition (mean bias, slope bias, residual error; Table 4). Prediction accuracy was assessed using the CCC thresholds suggested by Proctor et al. (2024): <0.40 poor, 0.40 to 0.60 moderate, and >0.60 moderate to high. Top- and bottom-performing published equations were identified, and the top-performing equa-

Table 2. Description of published BW prediction equations

Equation	Model type	Description ¹	Reference	Criterion ²
1	Linear	BW (kg) = 23.994 + 0.00271 × HC (cm)	Long et al. (2012)	—
2	Cubic	BW (kg) = 19.843 + 0.0732 × HC (cm) + 0.0001 × HC (cm) ² − 8 × EXP [−8 × HC (cm) ³]	Long et al. (2012)	—
3	Quadratic	BW (kg) = 71.128 − 2.28 × HG (cm) + 0.025 × HG (cm) ²	Silva et al. (2024)	—
4	Quadratic	BW (kg) = 58.459 − 1.979 × HG (cm) + 0.023 × HG (cm) ²	Silva et al. (2024)	—
5	Quadratic	BW (kg) = 62.449 − 1.979 × HG (cm) + 0.026 × HG (cm) ² − 0.284 × HH (cm) + 0.01 × HG (cm) ² − 0.011 × HG (cm) × HH (cm)	Silva et al. (2024)	—
6	Quadratic	BW (kg) = 14.756 − 0.946 × HG (cm) + 0.017 × HG (cm) ²	Silva et al. (2024)	—
7	Linear	LBW (kg) = −250.77 + 3.23 × HG (cm)	Bahashwan (2014)	—
8	Quadratic	LBW (kg) = −171.98 + 1.92 × HGC (cm) + 0.0054 × HGC (cm) × HGC (cm)	Bahashwan (2014)	—
9	Cubic	LBW (kg) = −171.98 + 1.92 × HG (cm) + 0.0054 × HG (cm) × HG (cm) + 0 × HG (cm) × HG (cm) × HG (cm)	Bahashwan (2014)	1
10	Exponential	LBW (kg) = EXP [2.051 + 0.024 × HGC (cm)]	Bahashwan (2014)	—
11	Exponential	LBW (kg) = 7.77 × 1.024 ^{HGC} (cm)	Bahashwan (2014)	—
12	Exponential	LBW (kg) = −1.722.48 + 388.86 × LN(HGC) (cm)	Bahashwan (2014)	—
13	Exponential	LBW (kg) = EXP [7.74 − 340.30/HGC (cm)]	Bahashwan (2014)	—
14	Exponential	LBW (kg) = 7.77 × EXP [0.024 × HGC (cm)]	Bahashwan (2014)	—
15	Exponential	LBW (kg) = 526.27 + −46.221.85/HGC (cm)	Bahashwan (2014)	—
16	Exponential	LBW (kg) = 0.00016 × HGC (cm) ^{2.85}	Bahashwan (2014)	—
17	Exponential	LBW (kg) = 1/0 + 0.129 × 0.98 ^{HGC} (cm)	Bahashwan (2014)	—
18	Linear	BW (kg) = −473 + 5.21 × HG (cm)	Ozkaya and Bozkurt (2009)	2
19	Linear	BW (kg) = −869 + 7.27 × HG (cm)	Ozkaya and Bozkurt (2009)	2
20	Linear	BW (kg) = −935 + 7.69 × HG (cm)	Ozkaya and Bozkurt (2009)	2
21	Linear	BW (kg) = 1.01 × partial weight scale (kg) − 1.21	Cantor et al. (2020)	1
22	Quadratic	BW (kg) = 407.8921916 − 9.2239388 × HG (cm) + 0.0699841 × HG (cm) ²	Heinrichs et al. (2017)	—
23	Quadratic	BW (kg) = 318 − 7.8648 × HG (cm) + 0.06497 × HG (cm) ²	Heinrichs et al. (1992)	—
24	Quadratic	BW (kg) = 277.3812486 − 7.1637099 × HG (cm) + 0.0610770 × HG (cm) ²	Heinrichs et al. (1992)	—
25	Cubic	BW (kg) = 65.36 − 1.966 × HG (cm) + 0.01959 × HG (cm) ² + 0.00001691 × HG (cm) ³	Heinrichs et al. (1992)	—
26	Linear	LW (kg) = −437.52 + 4.81 × HG (cm)	Tebug et al. (2016)	—
27	Linear	LW (kg) = −384.48 + 4.39 × HG (cm)	Tebug et al. (2016)	—
28	Linear	LW (kg) = −357.56 + 4.25 × HG (cm)	Tebug et al. (2016)	—
29	Linear	LW (kg) = −543.33 + 5.47 × HG (cm)	Tebug et al. (2016)	—
30	Linear	LW (kg) = −470.17 + 5.00 × HG (cm)	Tebug et al. (2016)	—
31:35	Exponential	BW (kg) = EXP [0.695653374133486 + 0.0361554502029309 × HG (cm)]	Rocha-Silva et al. (2024)	—
36	Exponential	BW (kg) = 0.130729457326762 − 0.000508018941877292 × sex male − 0.00103032418246943 × WH (cm) + 8.54772148626004 ^{−06} × HG (cm) × WH (cm) − 1	Rocha-Silva et al. (2024)	1
37	Linear	BW (kg) = −456.13 + 4.98 × HG (cm)	Udoh et al. (2021)	2
38	Linear	BW (kg) = −264.02 + 3.57 × HG (cm)	Udoh et al. (2021)	2
39	Linear	BW (lb) = −90.9 + 9.48 × HC (cm)	Pruitt et al. (1995)	—
40	Cubic	BW (kg) = −3.862 + 76.014 × HG (cm) − 0.488837 × HG (cm) ² + 0.00109755 × HG (cm) ³	Reis et al. (2008)	2
41	Exponential	BW ^{0.262} (kg) = 0.95 + 0.022 × HG (cm)	Lesosky et al. (2013)	—
42	Linear	LW (kg) = −662.6 + 6.373 × HG (cm)	Yan et al. (2009)	2
43	Linear	BW (kg) = −58.36 + 0.602 × HG (cm)	Wilson et al. (1997)	—
44	Linear	BW (kg) = −132.99 + 1.160 × HG (cm)	Wilson et al. (1997)	—
45	Linear	BW (kg) = −285.26 + 1.881 × HG (cm)	Wilson et al. (1997)	—
46	Linear	HC (cm) = −55.11 + 5.05 × BW (kg)	Parish et al. (2009)	1

¹LBW = live BW; LW = live weight; HC = hoof circumference; HG = heart girth; HH = hip height; WH = wither height.

²Selection criteria of equation removal = (1) Equations composed only of variables of interest and for which measurements are available in the current dataset; (2) equations specifically developed for calves and heifers, regardless of their sex class.

tions were further validated using an external validation dataset (Tedeschi, 2006).

Model Development

Heart girth was selected as the initial predictor for individual testing because it has been demonstrated to have the strongest correlation with BW among commonly measured calf morphological traits (Davis et al., 1961; Bahashwan, 2014; Chico-Alcudia et al., 2022). Model types, including linear, quadratic, cubic, and exponential, were developed and tested using forward stepwise regression, with fixed predictors added sequentially (i.e., HG, HG + HH, HG + HH + HC) and followed the same generic structure, where Y is the observed dependent variable of interest (BW), β_0 is the intercept, β_1 , β_2 , and β_3 are the regression coefficients, and e in the exponential model represents Euler's number (~ 2.718):

Linear models

$$Y = \beta_0 + \beta_1 \times \text{HG},$$

$$Y = \beta_0 + \beta_1 \times \text{HG} + \beta_2 \times \text{HH}, \text{ and}$$

$$Y = \beta_0 + \beta_1 \times \text{HG} + \beta_2 \times \text{HH} + \beta_3 \times \text{HC};$$

Quadratic model

$$Y = \beta_0 + \beta_1 \times \text{HG} + \beta_2 \times \text{HG}^2;$$

Cubic model

$$Y = \beta_0 + \beta_1 \times \text{HG} + \beta_2 \times \text{HG}^2 + \beta_3 \times \text{HG}^3; \text{ and}$$

Exponential model

$$Y = \beta_0 \times e^{\beta_1 \times \text{HG}}.$$

The selection of linear, polynomial (quadratic and cubic), and exponential model forms was guided by biological reasoning and prior literature on neonatal calf morphology (Wilson et al., 1997; Bahashwan, 2014; Chico-Alcudia et al., 2022). MacNeil et al. (2021) reported that lifetime cattle growth follows a sigmoidal trajectory; the neonatal period represents a narrow segment in which the relationship between BW and morphological traits may exhibit localized nonlinearity. Within this restricted range, incremental changes in girth or height can have disproportionate effects on BW, making polynomial and exponential functions biologically plausible approximations (Gibson et al., 2022). Exponential models were included because BW may respond proportionally rather

Table 3. Descriptive statistics of the dataset used to evaluate published equations to predict BW, and test model parameters

Item ²	Day 1 ¹				
	n	Mean	SD	Maximum	Minimum
Age (d)	—	1.0	0.0	—	—
Initial BW (kg)	320	42.6	5.6	61.7	31.8
HG (cm)	320	80.8	4.0	92	70
HH (cm)	320	77.5	3.6	89.5	68
HC (cm)	320	17.1	0.9	19.5	15
Independent dataset ³					
Age (d)	—	1.0	0.0	—	—
Initial BW (kg)	176	42.7	5.0	57.8	30.8
HG (cm)	176	79.7	3.5	88.9	66.0

¹Day = time point at which BW and morphological measures were collected.

²HG = heart girth; HH = hip height; HC = hoof circumference.

³Independent datasets = descriptive statistics used for the validation process.

than additively to changes in circumference-based traits, and prior studies have reported that exponential functions can describe the morphological measurement-BW relationship in young calves (Bahashwan, 2014; Chico-Alcudia et al., 2022). Testing multiple model structures enabled us to empirically determine which functional form best characterizes the BW-morphology association in day-old calves. Furthermore, previous studies (Wilson et al., 1997; Yan et al., 2009; Bahashwan, 2014) have demonstrated that simple linear models can reliably predict neonatal BW; therefore, the linear model was selected as the primary model for subsequent model validation. Assumptions of linear regression were evaluated using the Durbin–Watson and Breusch–Pagan test from the *car* package (Fox and Weisber, 2019), whereas the remaining diagnostic procedures including observed versus predicted plots for linearity, residual sequence plots for independence, residuals versus fitted plots for homoscedasticity, quantile-quantile plots and Shapiro–Wilk test, and Cook's distance for influential (Der and Everitt, 2002; R Core Team, 2024).

Linear mixed-effects models, including dairy farm as a random effect, were evaluated to assess whether clustering among the 14 source dairy farms influenced the regression coefficient or predictive performance. Animal was not included as a random effect, as each calf contributed with one BW observation. Because statistical testing is not used to evaluate random effects in this study, comparisons between fixed- and mixed-effects structures were based on changes in model coefficients and fit statistics. Linear mixed-effects models produced coefficient estimates and significance patterns that closely matched those from the fixed-effects linear model, with only a slight improvement in RMSE and CCC (Table 5

Table 4. Evaluation of BW predictions from published equations evaluated using the development dataset (n = 320) for day-old calves

Equation	Predicted mean (kg)	RMSE ¹ (% mean)	RMSE (kg)	Mean bias ² (% MSE)	Slope bias (% MSE)	Residual error (% MSE)	Mean bias	Slope bias	CCC ³
1	24.04	45.43	19.33	91.70	3.42	4.88	18.5	12.78	0.0001
2	34.67	21.31	9.07	75.55	2.03	22.42	7.88	1.67	0.1631
3	50.49	21.93	9.33	72.38	10.78	16.84	-7.94	9.39	0.3990
4	49.06	19.05	8.11	64.26	13.44	22.30	-6.50	8.84	0.4646
5	46.15	18.75	7.98	20.27	50.64	29.09	-3.59	32.23	0.5073
6	50.03	21.17	9.01	68.90	13.01	18.09	-7.48	10.56	0.4220
7	10.15	79.54	33.85	91.65	7.07	1.29	32.41	80.97	0.0841
8	18.44	59.83	25.46	89.67	8.06	2.27	24.11	52.24	0.1231
9	—	—	—	—	—	—	—	—	—
10	54.29	29.16	12.41	89.51	0.96	9.53	-11.74	1.48	0.2174
11	53.02	26.30	11.19	87.44	0.85	11.71	-10.47	1.06	0.2478
12	-15.20	140.79	59.91	92.92	6.67	0.41	57.75	239.45	0.0419
13	34.56	22.10	9.41	72.33	11.10	16.57	8.00	9.82	0.3975
14	54.05	28.60	12.17	89.14	0.96	9.91	-11.49	1.42	0.2233
15	-47.37	219.47	93.39	92.70	7.12	0.17	89.92	621.46	0.0259
16	43.93	10.86	4.62	8.90	22.36	68.74	-1.38	4.77	0.7035
17	39.78	11.28	4.80	33.52	2.82	63.67	2.78	0.65	0.5334
18	-52.13	226.25	96.28	96.72	3.12	0.16	94.69	289.51	0.0180
19	-281.73	764.40	325.29	99.38	0.61	0.01	324.28	642.16	0.0022
20	-313.80	839.84	357.40	99.42	0.57	0.01	356.36	731.08	0.0020
21	—	—	—	—	—	—	—	—	—
22	54.70	29.95	12.74	90.86	0.02	9.12	-12.15	0.03	0.1615
23	48.85	17.43	7.42	72.09	1.18	26.73	-6.30	0.65	0.4174
24	44.56	10.40	4.43	20.61	4.42	74.97	-2.01	0.86	0.6737
25	43.68	10.64	4.53	6.17	22.38	71.45	-1.12	4.59	0.7104
26	-48.97	218.28	92.89	97.08	2.75	0.17	91.52	237.11	0.0179
27	-29.85	173.41	73.80	96.28	3.45	0.27	72.41	187.71	0.0257
28	-14.24	137.29	58.42	94.51	5.05	0.43	56.80	172.53	0.0391
29	-71.46	271.42	115.50	97.44	2.45	0.11	114.02	326.37	0.0132
30	-66.27	258.69	110.09	97.72	2.16	0.12	108.82	261.35	0.0133
31:35	37.60	15.11	6.43	59.41	5.12	35.48	4.96	2.12	0.5195
36	—	—	—	—	—	—	—	—	—
37	858.42	1917.56	816.03	99.96	0.04	0.00	-815.86	257.93	0.0002
38	24.37	50.04	21.30	72.95	23.79	3.26	18.19	107.88	0.2046
39	71.50	69.79	29.70	94.94	2.98	2.07	-28.94	26.31	0.0661
40	-336.54	906.50	385.77	96.57	3.42	0.01	379.10	5,085.62	0.0040
41	46.29	13.19	5.61	44.31	9.09	46.60	-3.74	2.86	0.5969
42	-147.79	450.27	191.62	98.68	1.28	0.04	190.34	471.56	0.0056
43	-9.73	123.26	52.45	99.37	0.09	0.54	52.29	2.59	0.0071
44	-39.28	192.53	81.93	99.77	0.01	0.22	81.84	0.42	0.0056
45	-133.31	413.45	175.95	99.91	0.04	0.05	175.87	12.67	0.0020
46	—	—	—	—	—	—	—	—	—

¹RMSE = root mean square error.²MSE = mean squared error.³CCC = concordance correlation coefficient.

and Supplemental Table S1, see Notes). The small added gain and the minimal variance attributed to the dairy farm indicate that the random effect contributed little explanatory value for the BW estimates, supporting the use of linear fixed-effects models as the primary analytical approach.

Model Evaluation

All model types (i.e., linear, quadratic, cubic, and exponential) were evaluated using the *lme4* package (Bates et al., 2015), and CCC was calculated in the development dataset using the *epiR* package (Stevenson and Sergeant, 2025). Body weight prediction equations were evaluated

using custom R functions in combination with *base* R to compute RMSE, mean, and slope bias (kg and % MSE), and observed and predicted BW means (kg). Moreover, fit statistics (AIC and log-likelihood) were extracted directly from the model objects generated by *lme4*, but the error-based evaluation metrics were computed independently. Variance, covariance, and correlation matrices were extracted using the *base*, *stats*, and *utils* packages (R Core Team, 2018) and are presented in Supplemental Table S5 (see Notes).

Model Validation

Two complementary validation approaches were used to assess different components of model performance.

Table 5. Parameter estimates and fit statistics for BW predictions derived by forward selection using linear regression for day-old calves (n = 320)

Item ¹	Equation 1			Equation 2			Equation 3		
	Estimate	SE	P-value	Estimate	SE	P-value	Estimate	SE	P-value
Intercept	-41.1383	4.2066	<0.01	-54.8097	4.5857	<0.01	-60.8457	4.5185	<0.01
HG	1.0346	0.0520	<0.01	0.7898	0.0638	<0.01	0.6284	0.0677	<0.01
HH	—	—	—	0.4317	0.0715	<0.01	0.3585	0.0696	<0.01
HC	—	—	—	—	—	—	1.4460	0.2621	<0.01
Adjusted R ²	0.5575			0.6011			0.6354		
AIC	1,717.73			1,684.921			1,657.495		
Log-likelihood	-855.865			-838.4605			-823.7476		
Observed mean	42.4353			42.4353			42.4353		
Predicted mean	42.4353			42.4353			42.4353		
RMSE (% mean)	8.5700			8.1108			7.7418		
RMSE (kg)	3.6425			3.4528			3.3010		
Mean bias (% MSE)	0.00			0.00			0.00		
Slope bias (% MSE)	8.11			7.60			7.15		
Residual error (% MSE)	91.89			92.40			92.85		
Mean bias	0			0			0		
Slope bias	1.0694			0.8076			0.7691		
CCC	0.7159			0.7529			0.7797		

¹HG = heart girth; HH = hip height; HC = hoof circumference; AIC = Akaike information criterion; RMSE = root mean square error; MSE = mean squared error; CCC = concordance correlation coefficient.

The cross-evaluation quantified internal robustness by quantifying how stable model accuracy remained under repeated random partitioning of the development dataset (Yilmaz Adkinson et al., 2024). However, internal resampling cannot evaluate performance in a biologically distinct population (Aboshady et al., 2024). Therefore, an external independent dataset was used to validate generalizability across broader animal variation. Utilizing both methods enables a more comprehensive assessment of model performance (Tedde et al., 2021).

Internal Model Cross-Evaluation

Cross-evaluation was performed using 1,000 random simulations. Following Tedeschi (2015), 3 developed linear models were validated to predict BW in day-old calves. The development dataset was randomly split into 2 subsets for each simulation: 80% of the observations were allocated to the training set, and the remaining 20% to the testing set. All 3 linear models were fitted using ordinary least squares using the train function in the *caret* package (Kuhn, 2008). Cross-evaluation was quantified using independent adequacy statistics: RMSE, mean and slope bias, CV, prediction ratio, and CCC (Lin, 1989).

External Model Validation

An external and independent dataset described by Pereira et al. (2024) was used to further validate top-performing published equations (Equations 16, 24, and 25), which were selected based on their superior predictive

performance relative to the total published equations, as reflected in Table 4. Moreover, the linear equation developed with HG as the sole predictor (Equation 1) was included for validation, as the independent dataset contained only HG measurements. From the validation dataset, only Angus × Holstein male calves with BW and HG measurements were selected for validation, ensuring consistency with the population and measurement structure used to develop the 3 linear prediction models. Pereira et al. (2024) analyzed both male and female calves; however, in the present study, the Angus × Holstein subset containing only male calves was used for external validation. Validation consisted of comparing observed versus predicted values in the external dataset using the same statistical procedures applied during the internal evaluation.

The external dataset contained 176 Angus × Holstein crossbred male calves enrolled at the same commercial calf-raising facility within 24 h of birth. Body weight was individually recorded using a digital scale (Tru-test digital scale, model S3, Datamars, Mineral Well, TX) upon arrival, and HG was measured manually twice while calves stood square on the hutch floor, and the 2 measurements were averaged to obtain a single HG value for each calf.

RESULTS AND DISCUSSION

This study aimed to evaluate the accuracy and effectiveness of published equations and to develop, test, evaluate, cross-evaluate, and validate new models for estimating BW in day-old B × D male calves.

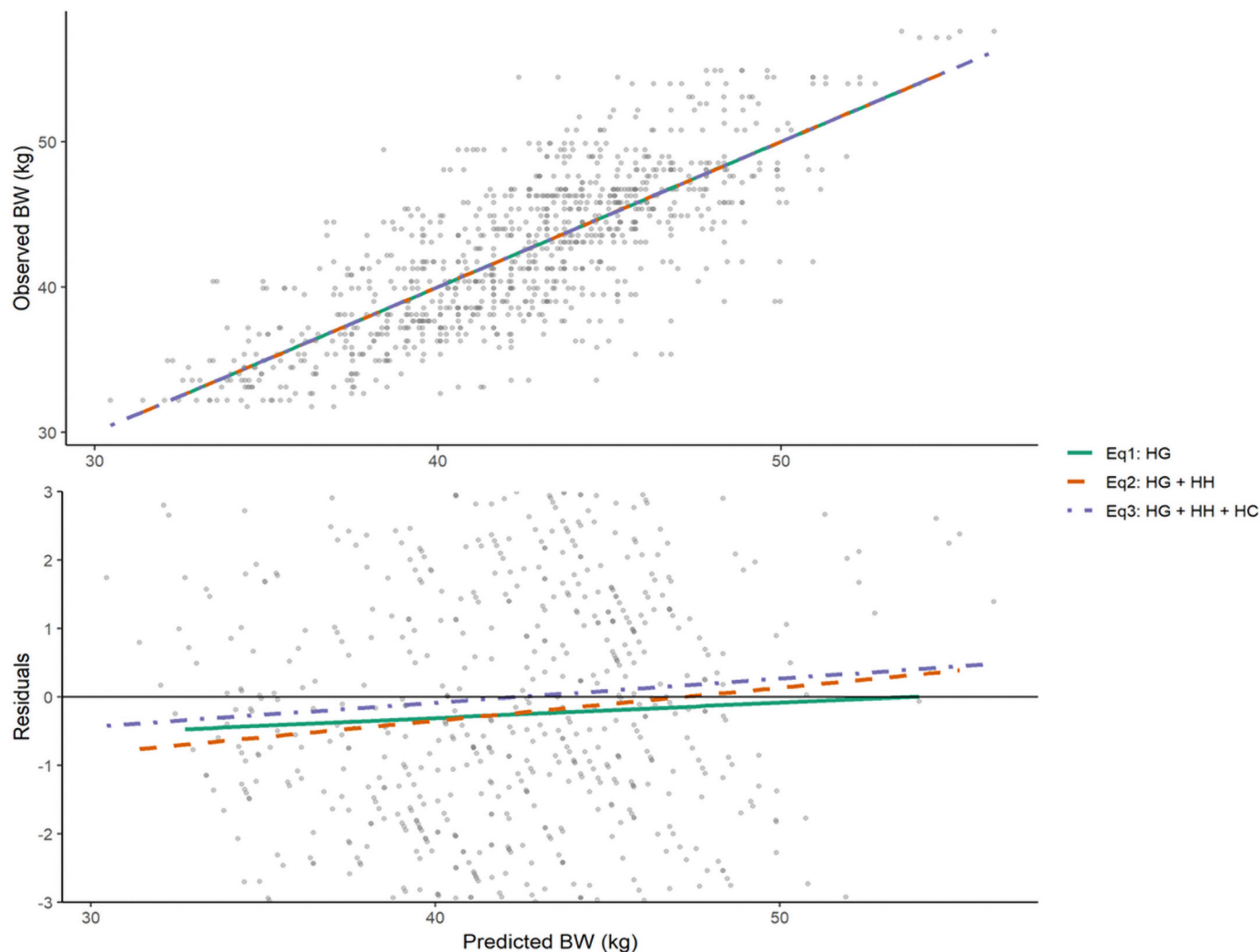


Figure 1. Predicted BW generated by linear equations Equation 1 (HG, solid green), Equation 2 (HG + HH, dashed orange), and Equation 3 (HG + HH + HC, dashed purple), using the development dataset. Residuals are plotted against predicted BW to evaluate model fit and error structure across equations.

Published Models

The top-performing published equations were Equations 16, 24, and 25 (Tables 2 and 4). Equation 25 predicted BW with the second least RMSE (4.53 kg) and the greatest CCC (0.71) among the top-performing published equations. Equation 16 had the largest RMSE (4.62 kg) and the second greatest CCC (0.70). Moreover, Equation 24 indicated a slightly smaller RMSE (4.43 kg) and CCC (0.67) compared with the other top-performing published equations. Overall, top-performing published equations indicated a robust predictive ability suitable for practical implementation to predict BW of day-old B $\times$ D male calves, which is consistent with Davis et al. (1961) and Heinrichs et al. (1992, 2007), who reported that BW can

be estimated using morphological measurements, such as HG, in preweaning dairy calves.

Model Development and Evaluation

Four model types (i.e., linear, quadratic, cubic, and exponential) were fitted and evaluated to predict BW in day-old B $\times$ D male calves. The model specifications, parameter estimates, and fit statistics of the quadratic, cubic, and exponential models are provided in Supplemental Tables S2, S3, and S4 (see Notes). Linear models with sequentially added morphometric predictors outperformed quadratic and cubic models. Cubic models did not yield significant parameter estimates for the intercept or predictor coefficients, and the exponential models did

Table 6. Cross-evaluation parameter estimates (n = 1,000) from developed equations

Item ¹	Equation 1 estimate	Equation 2 estimate	Equation 3 estimate
Observed mean (kg)	42.72	42.72	42.72
Predicted mean (kg)	43.20	43.11	43.02
SD for predicted	4.38	4.55	4.61
Mean SE	0.56	0.58	0.58
SD of SE	0.39	0.41	0.41
Mean CV	0.01	0.01	0.01
SD of CV	0.01	0.01	0.01
Mean predicted	1.01	1.01	1.01
SD of predicted	0.39	0.41	0.41
RMSE (kg)	3.96	3.78	3.53
Mean bias (% MSE)	0.00	0.00	0.73
Slope bias (% MSE)	1.48	1.06	0.14
Residual error (% MSE)	98.52	98.94	99.13
Mean bias	-0.48	-0.39	-0.30
Slope bias	0.55	0.59	0.62
CCC	0.7073	0.7397	0.7754

¹RMSE = root mean square error; MSE = mean squared error; CCC = concordance correlation coefficient.

not improve predictive performance, yielding RMSE and CCC values comparable to linear models. Therefore, 3 developed linear equations were selected, where Equation 1 predicts BW as a function of HG, Equation 2 as a function of HG and HH, and Equation 3 as a function of HG, HH, and HC (Table 5).

Model-predictor estimates (HG, HH, and HC) for developed linear equations were significant ($P < 0.01$) in predicting the BW of day-old B × D male calves (Table 5). Equation 1 with a single predictor variable (HG) had an RMSE of 3.64 kg and a CCC of 0.72. When a second predictor variable was added (HG and HH) to the model (Equation 2), the RMSE decreased by 5.40% (RMSE = 3.45 kg) and the CCC increased by 5.20% (CCC = 0.75) compared with Equation 1. Moreover, when a third predictor (HG, HH, and HC) was added (Equation 3), the RMSE decreased by 9.70% (RMSE = 3.30 kg) and the CCC increased by 8.90% (CCC = 0.78) compared with Equation 1. The incremental improvement in RMSE and CCC observed when HH and HC were added to the model was expected, as additional morphological predictors typically reduce unexplained variance in BW estimation models. Previous studies have demonstrated that calf BW is strongly associated with multiple body measurements, such as HG, HH, and HC (Heinrichs et al., 1992; Pruitt et al., 1995; Silva et al., 2024). Thus, the improvements observed are consistent with established morphological relationships.

Residual error accounted for most of the decomposition of prediction error across all models, ranging from 91.90% to 92.90% of the total MSE. Slope bias accounted for a small proportion of the total error, contributing 8.11%, 7.60%, and 7.15% for Equations 1, 2, and 3, respectively. The mean bias was the smallest component

Table 7. Validation of BW prediction equations using an independent dataset

Item ¹	N	SD	Observed mean (kg)	Predicted mean (kg)	RMSE ² (% mean)	RMSE (kg)	Mean bias ³ (% MSE)	Slope bias (% MSE)	Residual error (% MSE)	Mean bias	Slope bias	CCC ⁴
Equation 1	176	5.0	42.73	41.27	8.89	3.80	14.82	0.04	85.14	1.46	0.01	0.6372
Equation 16	176	5.0	42.73	42.14	9.01	3.85	2.37	16.99	80.64	0.59	2.52	0.7132
Equation 24	176	5.0	42.73	43.10	8.09	3.46	1.12	1.02	97.86	-0.37	0.12	0.7052
Equation 25	176	5.0	42.73	41.88	8.97	3.83	4.92	14.55	80.53	0.85	2.14	0.7115

¹Equation 1: developed BW prediction equation using heart girth (HG) only; Equation 16: Bahashwan, 2014; Equation 24: Heinrichs et al., 1992; Equation 25: Heinrichs et al., 1992.

²RMSE = root mean square error.

³MSE = mean squared error.

⁴CCC = concordance correlation coefficient.

of the total MSE, indicating that systematic offset errors were relatively small compared with the other sources of variance. Although equation 3 yielded the smallest RMSE and the greatest CCC among the 3 developed equations, it requires three morphological measurements per calf, thereby increasing labor demand during routine newborn calf processing. Moreover, HG (the common predictor among all 3 equations) can be collected with minimal restraint, whereas HH and HC require additional positioning and handling of the calf. Heinrichs et al. (2017) and Tebug et al. (2016) stated that HG-based models were preferred because they are easier to collect during newborn calf processing than other morphological measurements. Therefore, from a practical standpoint, Equation 1 would be most suitable for calf producers who require rapid BW estimates and have limited access to BW scales or time for multiple measurements.

Figure 1 demonstrates the observed versus predicted BW and residuals plotted against predicted BW for the 3 linear equations. The upper section of the figure reported the relationship between observed and predicted BW values for Equations 1, 2, and 3. All 3 developed equations reported strong linear alignment with minimal dispersion around the fitted lines. Meanwhile, the bottom section represented the residual distribution across the predicted BW range. Residuals were centered around zero for all models, with a reduced dispersion in Equations 2 and 3. Residual patterns did not exhibit systematic deviations across the range of predicted values, supporting model adequacy.

Internal Model Cross-Evaluation

Cross-evaluation results demonstrated that Equations 1, 2, and 3 exhibit similar predictive performance, indicating that all 3 models maintained consistent reliability across the resampling process (Table 6). Overall, all 3 developed equations overestimated BW (42.72 kg observed BW vs. 43.11 kg predicted BW). Equation 1 had an RMSE of 3.96 kg and a CCC of 0.71. When HH was added to Equation 2, the RMSE decreased to 3.78 kg (4.5% reduction) and the CCC increased to 0.74 (4.6% improvement) compared with Equation 1. When HC was added to Equation 3, the RMSE decreased to 3.53 kg (10.8% reduction) and the CCC increased to 0.78 (9.6% improvement) compared with Equation 1. Error decomposition showed that residual error dominated the total MSE for all models (98.52%–99.13%), indicating that prediction error was predominantly random. Consistent with these results, Tebug et al. (2016) reported that simple models using only HG achieved strong predictive performance (RMSE = 34.25 kg and $R^2 = 0.84$), with slight improvement when additional variables were included (RMSE = 32.81 kg and $R^2 = 0.86$).

Model Validation—External Dataset

A model validation test of Equation 1 was performed using an independent dataset containing 176 Angus × Holstein crossbred male calves (Table 7). Although 3 models were developed with HG, HH, and HC, only HG was recorded in the independent dataset; therefore, Equation 1 was the only equation eligible for external validation. In addition to this limitation, Equation 1 was retained as the primary equation because the internal cross-evaluations demonstrated that RMSE and CCC values were only marginally different from Equations 2 and 3, supporting its use as the primary equation.

When validating Equation 1, it was reported that the observed mean BW was 42.73 kg, the predicted mean was 41.27 kg, the RMSE was 3.80 kg, and the CCC was 0.64 (Table 7). In addition, mean and slope biases of 14.82% and 0.04%, respectively, were reported, with a residual error of 85.14%. Compared with the top-performing published equations evaluated using the same independent dataset (Equations 16, 24, and 25), Equation 1 demonstrated similar predictive accuracy, with published equations resulting in RMSE values ranging from 3.46 to 3.85 kg and CCC values ranging from 0.70 to 0.71. Overall, the validation results support existing evidence that morphological traits can reliably predict day-old BW. Using fewer predictors such as Equation 1 provide a practical, scalable tool for early-life BW estimation in settings where access to scales is limited.

CONCLUSIONS

The study demonstrated that published BW equations perform inconsistently when applied to day-old B × D calves, underscoring the need for population-specific models. The 3 developed linear equations improved predictive accuracy, with Equation 1 requiring only a single morphological measurement and maintaining a strong performance during external validation. These findings support the implementation of simple, empirically calibrated BW prediction models tailored to B × D calves to improve early-life management decisions in crossbred systems.

NOTES

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Nonstandard abbreviations used: AIC = Akaike information criterion; ASCR = Angus, Simmental, and Crossbred; B = both female and male; B × D = beef × dairy; BE = beef animals; BO = both dairy and beef animals; C = Curraleiro Pé-Duro; CCC = concordance correlation coefficient; D = Dhafari; DA = dairy animals; F = female; H = Holstein; HC = hoof circumference; HG = heart girth; HH = hip height; HJ = Holstein and Jersey; IZ = Indigenous Zebu; LBW = live BW; LW = live weight; M = male; MSE = mean square error; MU = multiple breeds not explicitly defined; RMSE = root mean squared error.

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